SOFTWARE REQUIREMENTS SPECIFICATION

Multi-Agent Financial Research System

An AI-Powered, Source-Grounded Financial Document Intelligence Platform

Prepared for: Infosys Springboard Virtual Internship | Sponsor: Vidzai Digital

Document Type: Software Requirements Specification (SRS)

Prepared in accordance with IEEE Std 830-1998

Version 1.0 | July 2026

Document Control

Revision History

Version	Date	Author	Description of Change
0.1	Week 2	Product / Engineering Team	Initial SRS draft derived from the approved PRD
1.0	Week 2	Product / Engineering Team	Finalized SRS, circulated for team and mentor review

Related Documents

- Product Requirements Document (PRD) — Multi-Agent Financial Research System, v1.0
- Vidzai Digital — Project Description for Infosys Interns

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional, non-functional, interface, and data requirements for the Multi-Agent Financial Research System — an AI-powered platform that allows users to upload company financial documents and receive source-grounded, cited answers, structured metric extraction, risk detection, multi-company comparison, and downloadable analyst-style PDF reports, produced by a coordinated pipeline of six specialized AI agents.

This document is written in accordance with the structure recommended by IEEE Std 830, and is intended to serve as the authoritative technical reference for the development team, QA team, and project stakeholders throughout the Infosys Springboard internship engagement. It complements, and should be read alongside, the Product Requirements Document (PRD) already produced for this project, which covers product vision, scope, personas, and milestones in greater business detail.

1.2 Document Conventions

- Requirement identifiers follow the pattern SRS-<MODULE>-<NUMBER>, for example SRS-AUTH-01.
- The keywords "shall" and "must" indicate mandatory requirements; "should" indicates a strong recommendation; "may" indicates optional behavior.
- Priority is expressed as High, Medium, or Low, reflecting criticality to the Minimum Viable Product (MVP).
- Use cases are identified as UC-<NUMBER> and are cross-referenced to the functional requirements they satisfy.

1.3 Intended Audience and Reading Suggestions

Audience	Suggested Focus
Developers / Engineers	Sections 3 (System Features), 4 (Interfaces), 6 (Data Requirements), 9 (Use Cases)
QA / Test Engineers	Sections 3, 7 (Non-Functional Requirements), 9 (Use Cases), 10 (Traceability Matrix)
Project Sponsor / Mentors	Sections 1, 2 (Overall Description), 7, 11 (Appendices)
UI/UX Designers	Sections 2.3 (User Classes), 4.1 (User Interfaces), 9 (Use Cases)
New Team Members	Read sequentially from Section 1 through Section 9

1.4 Project Scope

The system enables a user to upload financial documents (annual reports, 10-K filings, balance sheets, income statements, cash flow statements, MD&A, and auditor reports) into a research workspace. On upload, a Document Agent parses and indexes the file; an Extraction Agent pulls structured financial metrics; a Red Flag Agent detects anomalies and risks; a Comparison Agent benchmarks multiple companies; a Research Agent answers natural-language questions using Retrieval-Augmented Generation (RAG) with exact page citations; and a Report Agent compiles all outputs into a downloadable PDF. The system guarantees that no financial claim is presented without traceability to an uploaded source document.

Out of scope for this release: real-time market data, regulatory filing APIs, financial forecasting, portfolio recommendations, multi-language reports, voice interaction, and native mobile applications. These are documented as Future Scope in the PRD.

1.5 References

- Product Requirements Document (PRD) — Multi-Agent Financial Research System, v1.0
- Vidzai Digital — Project Description for Infosys Interns, "Multi-Agent Financial Research System"
- Infosys Springboard Virtual Internship 7.0 — Program Guidelines
- IEEE Std 830-1998 — IEEE Recommended Practice for Software Requirements Specifications

2. Overall Description

2.1 Product Perspective

The Multi-Agent Financial Research System is a new, standalone, self-contained web application. It is not a modification of an existing system. It consists of a React/Next.js frontend, a FastAPI backend exposing REST APIs, a MongoDB application database, a ChromaDB vector store, and a LangChain-orchestrated multi-agent pipeline backed by a large language model (Gemini or OpenAI).

The system architecture is described in full in the PRD (Section 18) and is summarized here as a single pipeline: User → Document Upload → Document Processing → Vector Database → Multi-Agent AI System → Chat Answers, Comparisons, Red Flags, and PDF Report.

2.2 Product Functions

At a summary level, the system shall provide the following major functions. Each is elaborated as a full System Feature in Section 3.

- User registration, authentication, and profile management.
- Creation and management of isolated research workspaces.
- Upload and automated processing (parsing, cleaning, chunking, embedding, indexing) of financial PDF documents.
- Automated extraction of structured financial metrics.
- Automated detection of financial red flags and risk indicators.
- Multi-company financial comparison and benchmarking.
- Conversational, citation-backed question answering grounded in uploaded documents.
- Generation and download of a structured, professional PDF research report.
- Persistence and retrieval of chat history and previously generated reports.

2.3 User Classes and Characteristics

User Class	Technical Expertise	Primary Use
Finance Student	Low to Medium	Ask direct questions about a single company's financials
MBA Candidate	Medium	Compare two or more companies for coursework
Early-Career Analyst	Medium to High	Draft a first-pass research note flagging risks
Chartered Accountant / Business Analyst	High (domain), Low (technical)	Verify extracted metrics and citations against source documents
Academic Mentor / Evaluator	Medium	Assess correctness, completeness, and citation accuracy of the system

2.4 Operating Environment

- Client: modern evergreen web browsers (Chrome, Edge, Firefox, Safari) on desktop and laptop devices, minimum resolution 1280×720.
- Server: Python 3.10+ runtime for the FastAPI backend, deployed on Render or Railway.
- Frontend deployment: Vercel (Next.js).
- Database: MongoDB (self-hosted or Atlas); Vector store: ChromaDB (embedded or hosted).
- LLM access: Gemini API (preferred) or OpenAI API (optional), accessed over HTTPS.

2.5 Design and Implementation Constraints

- The system must be implementable by a five-person student team within an 8-week internship window.
- The multi-agent pipeline shall be implemented using LangChain in a sequential (non-graph) execution model for v1.0.
- Financial answers shall never be generated without being grounded in retrieved document chunks (hard constraint, not configurable).
- LLM API usage is subject to third-party rate limits and token costs, constraining prompt size and retrieval depth.
- The application shall operate within free or low-cost tiers of its chosen cloud services for the internship deployment.

2.6 Assumptions and Dependencies

- Uploaded documents are assumed to be primarily text-based (not purely scanned) PDF files in English.
- The system depends on continued availability and quota of the selected third-party LLM API.
- 3–4 seed company annual reports are assumed to be available and cleared for use in the demo environment.
- The system assumes a single-tenant, internship-scale user base rather than production-grade concurrent traffic.

3. System Features

This section describes each major system feature (functionality area) using a consistent format: Description, Priority, Stimulus/Response, and detailed Functional Requirements. This structure allows each feature to be independently designed, built, and tested by different sub-teams.

3.1 User Authentication

Attribute	Detail
Priority	High
Description	Allows a user to register, log in, and manage a basic profile. Every workspace, document, chat, and report is scoped to an authenticated user.
Stimulus / Response	User submits registration or login credentials; system validates and returns a session token, or a descriptive error.

Functional Requirements

ID	Requirement
SRS-AUTH-01	The system shall allow a new user to register using name, email, and password.
SRS-AUTH-02	The system shall validate that the email is unique before creating an account.
SRS-AUTH-03	The system shall hash passwords using a secure, industry-standard algorithm before storage.
SRS-AUTH-04	The system shall authenticate a user and issue a JWT session token upon successful login.
SRS-AUTH-05	The system shall reject login attempts with invalid credentials and return a descriptive error without revealing which field was incorrect.
SRS-AUTH-06	The system shall allow an authenticated user to view and update their profile (name, email).
SRS-AUTH-07	The system shall expire JWT tokens after a configurable session duration and require re-authentication.
SRS-AUTH-08	The system shall reject any API request to a protected endpoint that lacks a valid JWT.

3.2 Research Workspace Management

Attribute	Detail
Priority	High
Description	Allows a user to create, view, rename, and delete isolated research workspaces, each of which contains its own documents, chat history, extracted metrics, red flags, and generated reports.
Stimulus / Response	User creates or selects a workspace; system scopes all subsequent uploads, chats, and reports to that workspace.

Functional Requirements

ID	Requirement
SRS-WS-01	The system shall allow an authenticated user to create a new workspace with a user-provided name.
SRS-WS-02	The system shall list all workspaces owned by the logged-in user, ordered by most recently updated.
SRS-WS-03	The system shall allow a user to rename an existing workspace.
SRS-WS-04	The system shall allow a user to delete a workspace, cascading deletion to its documents, embeddings, chats, metrics, and reports.
SRS-WS-05	The system shall prevent a user from accessing a workspace they do not own.
SRS-WS-06	The system shall support at least one pre-loaded, read-only seed workspace populated with sample company documents.

3.3 Financial Document Upload

Attribute	Detail
Priority	High
Description	Allows a user to upload one or more financial PDF documents into a workspace, with validation, storage, and confirmation.
Stimulus / Response	User selects and submits a PDF file; system validates file type/size, stores the file, and confirms success or reports an error.

Functional Requirements

ID	Requirement
SRS-DOC-01	The system shall accept file uploads with a .pdf extension and valid PDF content.
SRS-DOC-02	The system shall enforce a maximum file size limit (recommended: 50 MB) and reject files exceeding it.
SRS-DOC-03	The system shall reject non-PDF files and corrupted PDFs with a descriptive error message.
SRS-DOC-04	The system shall associate every uploaded document with the workspace and user that uploaded it.
SRS-DOC-05	The system shall display upload progress and a success or failure confirmation to the user.
SRS-DOC-06	The system shall allow a user to delete a previously uploaded document, removing its associated chunks and embeddings.
SRS-DOC-07	The system shall automatically trigger the Document Processing Pipeline upon successful upload, without requiring a separate manual step.

3.4 Document Processing Pipeline (Document Agent)

Attribute	Detail
Priority	High
Description	Parses an uploaded PDF, cleans and chunks its text, generates embeddings, and stores them in the vector database with page-level metadata.
Stimulus / Response	Document Agent receives a stored PDF; outputs indexed, embedded chunks with preserved metadata.

Functional Requirements

ID	Requirement
SRS-PARSE-01	The Document Agent shall extract raw text from every page of the uploaded PDF using a PDF parsing library (e.g., PyMuPDF).
SRS-PARSE-02	The Document Agent shall record the originating page number for every extracted text segment.
SRS-PARSE-03	The Document Agent shall apply OCR to pages with no extractable text layer, where OCR is enabled.
SRS-PARSE-04	The Document Agent shall clean extracted text by removing headers, footers, page numbers, and non-informative artifacts.
SRS-PARSE-05	The Document Agent shall chunk cleaned text into segments of a configurable target size with a defined overlap between adjacent chunks.
SRS-PARSE-06	The Document Agent shall generate a unique chunk ID for every chunk.
SRS-PARSE-07	The Document Agent shall generate an embedding vector for every chunk using the configured embedding model.
SRS-PARSE-08	The Document Agent shall store each embedding in the vector database together with document name, page number, section (if identifiable), and chunk ID.
SRS-PARSE-09	The Document Agent shall report processing status (queued, processing, complete, failed) back to the workspace.

3.5 Financial Metric Extraction (Extraction Agent)

Attribute	Detail
Priority	High
Description	Automatically identifies and extracts standard financial metrics from indexed document chunks and stores them as structured records.
Stimulus / Response	Extraction Agent receives indexed chunks of a document; outputs a structured financial metrics record.

Functional Requirements

ID	Requirement
SRS-EXT-01	The Extraction Agent shall extract Revenue, Net Profit, EBITDA, and EPS where disclosed in the document.
SRS-EXT-02	The Extraction Agent shall extract ROE, ROCE, Debt, Assets, and Liabilities where disclosed.
SRS-EXT-03	The Extraction Agent shall extract Cash Flow, Operating Margin, and Gross Margin where disclosed.
SRS-EXT-04	The Extraction Agent shall extract narrative market and business risk disclosures as supporting text.
SRS-EXT-05	The Extraction Agent shall store every extracted value with a reference to its source page.
SRS-EXT-06	The Extraction Agent shall mark a metric as "not found" rather than fabricating a value when it is not disclosed in the document.
SRS-EXT-07	The Extraction Agent shall associate extracted metrics with the correct reporting period when multiple years are present in a single document.

3.6 Financial Red Flag Detection (Red Flag Agent)

Attribute	Detail
Priority	High
Description	Automatically scans extracted metrics and document text for indicators of financial risk, without requiring the user to request it.
Stimulus / Response	Red Flag Agent receives extracted metrics for a document; outputs a list of warnings with severity and source reference.

Functional Requirements

ID	Requirement
SRS-FLAG-01	The Red Flag Agent shall detect a declining revenue trend across reported periods.
SRS-FLAG-02	The Red Flag Agent shall detect an increasing debt level across reported periods.
SRS-FLAG-03	The Red Flag Agent shall detect negative or sharply declining operating cash flow.
SRS-FLAG-04	The Red Flag Agent shall detect auditor qualifications or going-concern language in the auditor's report.
SRS-FLAG-05	The Red Flag Agent shall detect disclosed litigation or material legal risks.
SRS-FLAG-06	The Red Flag Agent shall detect high liabilities relative to assets and declining margins.
SRS-FLAG-07	The Red Flag Agent shall assign a severity level (Low, Medium, High) to every generated warning.
SRS-FLAG-08	The Red Flag Agent shall run automatically once the Extraction Agent completes, without user prompting.

3.7 Multi-Company Comparison (Comparison Agent)

Attribute	Detail
Priority	Medium
Description	Cross-references extracted financial data across two or more companies present in a workspace to produce a benchmarking table.
Stimulus / Response	User selects two or more companies for comparison; Comparison Agent outputs a structured comparison table.

Functional Requirements

ID	Requirement
SRS-CMP-01	The Comparison Agent shall generate a side-by-side comparison of Revenue, Net Profit, and margins across selected companies.
SRS-CMP-02	The Comparison Agent shall generate a comparison of Debt and key financial ratios (ROE, ROCE) across selected companies.
SRS-CMP-03	The Comparison Agent shall indicate which company performs better or worse on each compared metric.
SRS-CMP-04	The Comparison Agent shall only run when two or more companies with extracted metrics exist in the workspace.
SRS-CMP-05	The Comparison Agent shall handle a missing metric for one company gracefully, marking it as unavailable rather than omitting the row.

3.8 Conversational Research (Research Agent / RAG)

Attribute	Detail
Priority	High
Description	Allows a user to ask natural-language financial questions and receive answers strictly grounded in uploaded documents, with exact citations, using Retrieval-Augmented Generation.
Stimulus / Response	User submits a question in the chat interface; Research Agent retrieves relevant chunks, generates a grounded answer, and returns it with citations.

Functional Requirements

ID	Requirement
SRS-RAG-01	The Research Agent shall accept free-text, single-part or multi-part financial questions.
SRS-RAG-02	The Research Agent shall convert the user's question into an embedding and retrieve the top-k most semantically similar chunks from the vector database.
SRS-RAG-03	The Research Agent shall generate an answer using only the retrieved chunks as grounding context.
SRS-RAG-04	The Research Agent shall reason step-by-step when a question contains multiple sub-parts.

ID	Requirement
SRS-RAG-05	The Research Agent shall attach the document name and exact page number as a citation for every factual claim.
SRS-RAG-06	The Research Agent shall explicitly state that it cannot find an answer when no sufficiently relevant chunk is retrieved, rather than generating an unsupported answer.
SRS-RAG-07	The Research Agent shall persist every question, answer, and citation set to the workspace's chat history.
SRS-RAG-08	The Research Agent shall return an answer within the performance targets defined in Section 7.1.

3.9 Report Generation (Report Agent)

Attribute	Detail
Priority	High
Description	Compiles the outputs of all other agents into a single, structured, professional PDF research report that the user can download.
Stimulus / Response	User requests report generation for a workspace; Report Agent compiles all prior agent outputs and renders a downloadable PDF.

Functional Requirements

ID	Requirement
SRS-RPT-01	The Report Agent shall compile an Executive Summary synthesizing the workspace's key findings.
SRS-RPT-02	The Report Agent shall include a Financial Overview and Key Financials section sourced from the Extraction Agent.
SRS-RPT-03	The Report Agent shall include a Risks section sourced from the Red Flag Agent.
SRS-RPT-04	The Report Agent shall include a Comparative Analysis section sourced from the Comparison Agent, when applicable.
SRS-RPT-05	The Report Agent shall include Strengths, Weaknesses, Financial Ratios, AI Insights, and a Final Conclusion section.
SRS-RPT-06	The Report Agent shall render the compiled report as a downloadable PDF file.
SRS-RPT-07	The Report Agent shall allow the user to regenerate the report after new documents are added or existing ones are updated.
SRS-RPT-08	The Report Agent shall complete generation within the performance target defined in Section 7.1.

3.10 Chat History and Report History

Attribute	Detail
Priority	Medium

Attribute	Detail
Description	Allows a user to revisit past conversations and previously generated reports within a workspace.
Stimulus / Response	User opens the History view; system displays past chats and reports for the selected workspace.

Functional Requirements

ID	Requirement
SRS-HIST-01	The system shall persist and display chat history in chronological order per workspace.
SRS-HIST-02	The system shall allow a user to re-download any previously generated report.
SRS-HIST-03	The system shall allow a user to search chat history by keyword within a workspace.

3.11 Account Settings

Attribute	Detail
Priority	Low
Description	Allows a user to manage basic account settings and securely end their session.
Stimulus / Response	User opens Settings; system displays editable account fields and a logout action.

Functional Requirements

ID	Requirement
SRS-SET-01	The system shall allow a user to update their display name and password.
SRS-SET-02	The system shall allow a user to log out, invalidating the current session token.

4. External Interface Requirements

4.1 User Interfaces

The system shall provide a responsive web interface built with React, Next.js, and Tailwind CSS, covering the following primary screens:

Screen	Purpose
Login / Register	Authenticate or create a new user account
Dashboard	List all workspaces belonging to the user; create a new workspace
Workspace View	Show uploaded documents, extracted metrics, and red flags for the selected workspace
Upload Panel	Upload new financial documents with progress and validation feedback
Chat Interface	Ask questions and view cited, grounded answers in a conversational layout
Comparison View	Select companies and view a side-by-side benchmarking table
Report View	Trigger report generation, preview, and download the PDF report
History	Browse past chat sessions and previously generated reports
Settings	Update profile information and log out

- The interface shall clearly display the source citation (document name and page number) alongside every AI-generated answer.
- The interface shall clearly distinguish between AI-generated content and raw extracted data.
- The interface shall show a loading/progress indicator during document processing and report generation.

4.2 Hardware Interfaces

The system has no direct hardware interface requirements. It is a cloud-hosted web application accessed via standard client devices (laptop or desktop computers) with a keyboard, display, and network connection. No specialized hardware, sensors, or peripherals are required.

4.3 Software Interfaces

Software Component	Interface Description
LLM Provider (Gemini API / OpenAI API)	HTTPS REST API used for answer generation, extraction reasoning, and report drafting
Embedding Model	HTTPS REST API or local library used to convert text chunks and queries into vector embeddings
MongoDB	Database driver (PyMongo / Motor) used for all application data (users, workspaces, documents, chats, metrics, reports, logs)
ChromaDB	Vector database client used to store and query document chunk embeddings

Software Component	Interface Description
PyMuPDF	Python library used to parse and extract text and page metadata from uploaded PDFs
Tesseract OCR (optional)	OCR engine invoked for scanned pages with no extractable text layer
ReportLab / WeasyPrint	PDF generation library used by the Report Agent to render the final report

4.4 Communication Interfaces

- All communication between the frontend and backend shall occur over HTTPS using JSON-formatted REST API requests and responses.
- All communication with third-party LLM and embedding APIs shall occur over HTTPS.
- JWT bearer tokens shall be transmitted in the Authorization header of every authenticated request.

5. System Architecture Overview

The full architecture, technology stack, and multi-agent orchestration flow are specified in detail in the accompanying Product Requirements Document (PRD), Sections 16 through 19. This SRS summarizes the architecture as a layered pipeline for requirement traceability purposes:

```
React / Next.js Frontend

↓  (REST over HTTPS)

FastAPI Backend

↓

MongoDB (application data)      ChromaDB (embeddings)

↓

LangChain Orchestration Layer

↓

Document → Extraction → Red Flag → Comparison → Research → Report Agents

↓

Chat Answers + Comparison Tables + Red Flags + PDF Report
```

6. Data Requirements

6.1 Logical Data Model

The system uses MongoDB as the primary application datastore, chosen for its flexible, document-oriented schema, which accommodates the variability of financial report structures. Vector embeddings are stored separately in ChromaDB. The logical entities and their relationships are summarized below.

Entity	Relationship	Description
User	1 : Many Workspace	A user owns zero or more workspaces
Workspace	1 : Many Document	A workspace contains zero or more uploaded documents
Document	1 : Many Chunk (in ChromaDB)	A document is split into many indexed chunks
Document	1 : 1 FinancialMetrics	A document has one structured metrics record
Document	1 : Many RedFlag	A document may have zero or more detected red flags
Workspace	1 : Many ChatMessage	A workspace accumulates a chronological chat history
Workspace	1 : Many Report	A workspace may have multiple generated report versions
Agent Execution	Many : 1 Workspace	Every agent run is logged against the workspace it operated on

6.2 Data Dictionary

Collection	Field	Description
Users	_id, name, email, passwordHash, createdAt	Registered user account
Workspaces	_id, userId, workspaceName, createdAt, updatedAt	A user's isolated research session
Documents	_id, workspaceId, filename, totalPages, status, uploadedAt	Metadata for an uploaded financial document
FinancialMetrics	documentId, revenue, netProfit, ebitda, eps, roe, roce, debt, assets, liabilities, cashFlow, operatingMargin, grossMargin	Structured metrics extracted from a document
RedFlags	documentId, flagType, description, severity, sourcePage	A single detected risk indicator

Collection	Field	Description
ChatHistory	workspaceId, userQuestion, aiAnswer, citations[], timestamp	One question/answer turn with citations
Reports	_id, workspaceId, reportUrl, generatedAt, version	A generated PDF report's metadata
AgentLogs	agentName, workspaceId, input, output, durationMs, timestamp, status	Execution log for a single agent run

Vector store (ChromaDB) records: chunkId, embedding vector, documentName, page, section, and a back-reference to the parent documentId.

7. Non-Functional Requirements

7.1 Performance Requirements

Requirement	Target
Single-document question response time	≤ 8 seconds
Multi-document comparison response time	≤ 15 seconds
Vector chunk retrieval latency	≤ 1 second per query
Document processing time (per 100 pages)	≤ 60 seconds
Report generation time	≤ 30 seconds

7.2 Security Requirements

- All passwords shall be stored using a one-way, salted hash; plaintext passwords shall never be stored or logged.
- All API endpoints except registration and login shall require a valid JWT bearer token.
- A user shall only be able to access workspaces, documents, chats, and reports that they own.
- All data in transit shall be encrypted using HTTPS/TLS.
- Uploaded files shall be scanned for valid PDF structure before processing to reduce injection risk.

7.3 Reliability and Availability

- The document processing pipeline shall successfully complete for at least 95% of valid, text-based PDF uploads.
- The application shall target 99% uptime during the demonstration and evaluation period.
- A failed agent execution shall be logged and retried at least once before being surfaced to the user as an error.

7.4 Scalability

- The system shall support at least 50 concurrent workspaces and 500 indexed documents without significant performance degradation in the internship demo environment.
- The vector database and orchestration layer shall be designed so that additional agents can be added without re-architecting existing components.

7.5 Maintainability and Extensibility

- Each of the six agents shall be implemented as an independent, testable module with a clearly defined input/output contract.
- Configuration values (chunk size, retrieval top-k, LLM model name) shall be externalized rather than hard-coded.

- The codebase shall follow the folder structure defined in the PRD (Section 19.4) to ensure consistency across the team.

7.6 Usability

- A first-time user shall be able to upload a document and receive an answered question within 3 minutes without external guidance.
- Error messages shall be written in plain language, avoiding raw stack traces or technical jargon.
- The interface shall provide visible confirmation for every user action (upload success, report generation complete, etc.).

7.7 Accuracy and Grounding (AI-Specific Requirement)

- The system shall never present a financial figure, claim, or citation that cannot be traced to a specific page in an uploaded source document.
- The Research Agent shall decline to answer, rather than guess, when retrieval confidence is low or no relevant chunk is found.
- Every report generated by the Report Agent shall be fully traceable to the underlying agent outputs and source documents used to compile it.

8. Other Requirements

8.1 Legal and Compliance

- Seed company documents used for demonstration shall be sourced from publicly available annual reports and used strictly for educational, non-commercial purposes.
- User-uploaded documents shall not be shared across workspaces or users without explicit action by the owning user.
- The system shall not be represented as providing investment advice; all outputs are for research and educational purposes only.

8.2 Internationalization

- The v1.0 release supports English-language financial documents and interface text only. Multi-language support is deferred to Future Scope.

8.3 Reuse Objectives

- The Document Agent's parsing and chunking pipeline shall be built as a reusable module independent of the specific financial-domain agents, enabling future reuse for non-financial document types.
- The orchestration layer shall expose a generic "register agent" interface so agents can be added or removed with minimal changes to the pipeline controller.

9. Use Case Specifications

The following use cases describe the system's primary interactions in detail, including actors, preconditions, main flow, alternate flows, and postconditions. Each use case is cross-referenced to the functional requirements in Section 3.

UC-01 — Register and Log In

Attribute	Detail
Actor	Prospective / Returning User
Precondition	User has a valid email address; for login, an account already exists.
Postcondition	All subsequent workspace and document operations are scoped to the authenticated user.
Related Requirements	SRS-AUTH-01 to SRS-AUTH-08

Main Flow

1. User navigates to the Register or Login screen.
2. User submits name, email, and password (registration) or email and password (login).
3. System validates the submitted data.
4. System creates the account (registration) or verifies credentials (login).
5. System issues a JWT session token and redirects the user to the Dashboard.

Alternate / Exception Flows

- If the email is already registered, the system displays "Account already exists" and suggests logging in instead.
- If credentials are invalid at login, the system displays a generic "Invalid email or password" message.

UC-02 — Upload a Financial Document

Attribute	Detail
Actor	Authenticated User
Precondition	User is logged in and has selected or created a workspace.
Postcondition	Document is stored, processed, indexed, and ready for extraction, red-flag detection, and question answering.
Related Requirements	SRS-DOC-01 to SRS-DOC-07, SRS-PARSE-01 to SRS-PARSE-09

Main Flow

6. User selects "Upload Document" within a workspace.
7. User selects a PDF file from their device.
8. System validates the file type and size.

9. System stores the file and confirms the upload.
10. System automatically triggers the Document Agent to parse, clean, chunk, embed, and index the document.
11. System updates the workspace to show the document as "Processed" and available for analysis.

Alternate / Exception Flows

- If the file is not a valid PDF, the system rejects the upload and displays an error.
- If the file exceeds the size limit, the system rejects the upload and displays an error.
- If parsing fails, the system marks the document as "Failed" and allows the user to retry.

UC-03 — Ask a Financial Question

Attribute	Detail
Actor	Authenticated User
Precondition	At least one document in the workspace has completed processing.
Postcondition	The question and its grounded, cited answer are saved to the workspace's chat history.
Related Requirements	SRS-RAG-01 to SRS-RAG-08

Main Flow

12. User opens the Chat interface for a workspace.
13. User types a natural-language financial question and submits it.
14. System (Research Agent) converts the question into an embedding and retrieves the most relevant chunks.
15. System generates an answer grounded strictly in the retrieved chunks.
16. System displays the answer along with the source document name and page number(s).
17. System saves the question, answer, and citations to chat history.

Alternate / Exception Flows

- If no sufficiently relevant chunk is found, the system responds that it cannot answer from the uploaded documents.
- If the question has multiple parts, the system answers each part with its own citation, using step-by-step reasoning.

UC-04 — Compare Multiple Companies

Attribute	Detail
Actor	Authenticated User
Precondition	At least two companies with processed documents and extracted metrics exist in the workspace.

Attribute	Detail
Postcondition	A benchmarking table is displayed and made available to the Report Agent.
Related Requirements	SRS-CMP-01 to SRS-CMP-05

Main Flow

18. User selects two or more companies within a workspace.
19. User selects "Compare."
20. System (Comparison Agent) retrieves extracted metrics for each selected company.
21. System generates a comparison table covering revenue, profit, margins, debt, and ratios.
22. System displays the table, highlighting the stronger or weaker performer per metric.

Alternate / Exception Flows

- If a metric is missing for one company, the system marks it as "Not Available" in the table rather than omitting the row.

UC-05 — Generate a Research Report

Attribute	Detail
Actor	Authenticated User
Precondition	At least one document has been processed, with extraction and red-flag detection complete.
Postcondition	A downloadable PDF report is available in the workspace's report history.
Related Requirements	SRS-RPT-01 to SRS-RPT-08

Main Flow

23. User selects "Generate Report" within a workspace.
24. System (Report Agent) collects outputs from the Extraction, Red Flag, Comparison, and Research agents.
25. System compiles the Executive Summary, Financial Overview, Risks, Comparative Analysis, and Conclusion sections.
26. System renders the compiled content as a PDF file.
27. System makes the PDF available for download and adds it to the workspace's report history.

Alternate / Exception Flows

- If the workspace contains only one company, the Comparative Analysis section is omitted or marked as not applicable.
- If report generation fails, the system logs the error and notifies the user with an option to retry.

UC-06 — Review Automatically Detected Red Flags

Attribute	Detail
Actor	Authenticated User
Precondition	A document has completed metric extraction.
Postcondition	The user has visibility into all detected risk indicators without needing to ask a question.
Related Requirements	SRS-FLAG-01 to SRS-FLAG-08

Main Flow

28. System (Red Flag Agent) runs automatically once the Extraction Agent completes for a document.
29. System evaluates extracted metrics and document text for known risk patterns.
30. System generates a list of warnings, each with a severity level and source page reference.
31. System displays the red flags in the workspace view without requiring the user to ask.

Alternate / Exception Flows

- If no red flags are detected, the system displays a clear "No significant red flags detected" message rather than leaving the section blank.

10. Requirements Traceability Matrix

This matrix maps each major use case to its owning agent(s) and functional requirement group, supporting test planning and coverage verification.

Use Case	Primary Agent(s)	Requirement Group	Test Focus
UC-01	Authentication Service	SRS-AUTH	Registration, login, session validation
UC-02	Document Agent	SRS-DOC, SRS-PARSE	Upload validation, parsing accuracy, page metadata
UC-03	Research Agent	SRS-RAG	Retrieval relevance, citation accuracy, no-hallucination
UC-04	Comparison Agent	SRS-CMP	Metric alignment across companies, missing-data handling
UC-05	Report Agent	SRS-RPT	Section completeness, PDF rendering, regeneration
UC-06	Red Flag Agent	SRS-FLAG	Detection accuracy, severity assignment, false positive rate

11. Appendices

11.1 Glossary

Term	Definition
RAG	Retrieval-Augmented Generation — retrieving relevant source content before generating an answer, to ground the response in fact.
Embedding	A numerical vector representation of text that captures semantic meaning, enabling similarity search.
Chunk	A segment of a document's text, sized appropriately for retrieval and embedding.
Vector Database	A database optimized for storing and searching embeddings by similarity (e.g., ChromaDB).
Red Flag	An indicator of financial risk or concern, such as declining margins or rising debt.
Orchestration Layer	The component that decides which agent runs, in what order, and what data passes between agents.
JWT	JSON Web Token — a signed token used to authenticate API requests.
MD&A	Management Discussion & Analysis — a narrative section of an annual report explaining financial results.

11.2 Analysis Models

Detailed architecture diagrams, the multi-agent orchestration flow, entity relationships, and the full technology stack are provided in the Product Requirements Document, Sections 16 through 19, and are incorporated here by reference to avoid duplication of maintained diagrams.

11.3 To Be Determined (TBD) List

TBD ID	Open Item	Owner
TBD-01	Final choice between Gemini API and OpenAI API for production LLM calls	Engineering Lead
TBD-02	Exact chunk size and overlap values to be tuned during Milestone 2 testing	Document Agent Owner
TBD-03	Final list of 3-4 seed companies to pre-load for demonstration	Product Team
TBD-04	Whether MongoDB Atlas Vector Search will replace the separate ChromaDB instance	Engineering Lead

11.4 Document End

This SRS is a living document maintained alongside the PRD. As implementation progresses, functional requirement details, data schemas, and use case flows should be revised to reflect actual engineering decisions, with changes tracked in a revision history.